SERENEDB CHALLENGE · NATIONAL AI HACKATHON 2026

Log Intelligence Agent

The right insights for the right people: one workflow from question to evidence to recommendation.

4,747,963
BGL log lines

207,820
OpenStack log lines

16
MCP tools Claude chooses from

THE PROBLEM

Millions of log lines don’t fit in a prompt

Engineer

What failed, in which component, and what happened just before and after?

Team lead

Which problems recur, how often, and which team owns them?

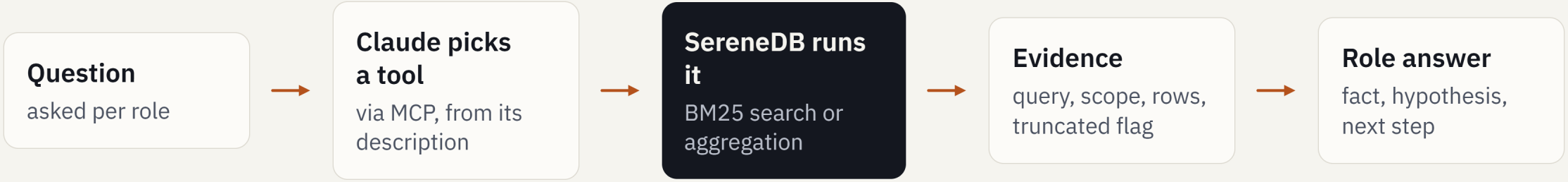
Manager

Which observed problems deserve attention first, and what happens next?

Same logs, three questions: each role needs **different evidence and a different action**, not just a shorter answer.

THE APPROACH

The database searches and counts; the agent explains



41 s

to load 4.7M BGL lines with SereneDB's server-side read_csv

BM25

inverted-index full-text search, with time filters

No free SQL

Claude chooses parameterized tools; results are capped

THE TOOLS

Sixteen tools, each with a clear job

Role	Tools Claude can call	Returns
Engineer	search_logs_fulltext · get_nearby_events · request_trace	BM25-ranked lines, context window
Team lead	alert_categories · alert_profile · count_alerts_by_team · trend_by_day · trend_by_week · top_nodes_for_alert	whole-corpus counts, spread, owners, trends
Manager	openstack_by_component · latency_stats · alert_profile	per-service rates, latency, owners
Any role	corpus_scope · team_ownership	data scope, component-to-team map
Huge data	otel_setup · otel_aggregate · otel_search	1B-row remote Parquet, in place

Every tool returns the same envelope: **scope, filters, the SQL that ran, rows, count, truncated**.

ENGINEER · LIVE CLAUDE SESSION

One chip error took down a whole job

QUESTION

“Investigate KERNRTSP alerts: which nodes and days, example lines, and what happened before and after the most recent one?”

SCOPE FOUND

3,983 lines on 3,394 nodes over 126 days; half from one mass event on 14 Jun 2005, then a steady trickle

MOST RECENT: 3 JAN 2006

R27-M0-NA-C:J11-U11

OBSERVED

02:24–03:24 cluster quiet, no warning

03:24:46 L2 d-cache parity machine check on that node

03:25:05 rts panic (KERNRTSP)

03:25:06+ APPSEV “link severed” on I/O nodes in six racks, all to one front-end

HYPOTHESIS

A hardware fault in that chip’s L2 cache ended one large job. Not a confirmed root cause.

NEXT STEP

Diagnose R27-M0-NA-C:J11-U11; R63-M0-NE-C:J12-U01 (22 panics, the most of any node) is the clearest replacement candidate.

Scope: BGL, 4,747,963 lines · full-text search, node and cluster time windows · 5

TEAM LEAD · LIVE CLAUDE SESSION

The loudest alert is not the recurring one

Category	Lines	Nodes	Days	Peak day	Owner
KERNDTLB	152,734	541	10	99.6%	Kernel Platform
KERNSTOR	63,491	22,529	4	100%	Kernel Platform
APPSEV	49,651	1,024	70	6.2%	App Runtime
KERNMNTF	31,531	1,024	26	13.8%	Kernel Platform
KERNRTSP	3,983	3,394	126	51.4%	Kernel Platform
KERNMC	342	265	144	2.9%	Kernel Platform

6 of 41 categories shown. Orange: burst (peak day ≥ 90% of lines). Blue: still active at the end of the data.

99.6%

of KERNDTLB's 152,734 lines fell on one day, 12 Jun 2005: a burst, not a recurring defect.

Most persistent: KERNMC machine checks (144 days, only 342 lines), KERNRTSP (126 days), APPSEV (70 days, all 1,024 nodes).

Scope: whole BGL corpus, all 41 alert categories, Jun 2005 – Jan 2006 · alert_profile, count_alerts_by_team, trend_by_day · 6

MANAGER · LIVE CLAUDE SESSION

Ongoing problems first, bursts as reviews

1	Application Runtime	APPSEV: 70 days, all 1,024 nodes, no day above 6%. Systemic, not a node swap; start with what changed mid-November
2	API Gateway	nova-api: flat 2.01% error lines every day (2,068 of 102,868); confirm what the 2% is, then the latency tail (p99 514 ms)
3	Kernel Platform	KERNMNTF and KERNTERM (26 days each); KERNMC (144 days) and KERNRTSP (126 days): inspect nodes in both top-5 lists first
4	Compute	nova-compute: 204 error lines (0.20%), one repeating image-cache job; low priority
↺	Post-incident reviews	One-day bursts: KERNDTLB, KERNSTOR (76% of Kernel Platform's lines), APPTO, APPOUT, APPBUSY

Ranked by measured persistence and volume within each dataset · owners from the ownership table · no invented cost, downtime or impact · 7

SHOWS ITS WORK

Every number comes with its query

```
[scope: BGL (4,747,963 lines) | period: 2006-01-03 to 2006-01-04
| method: BM25 full-text]

SELECT …, BM25(bgl_idx.tableoid) AS score FROM bgl_idx WHERE
(message @@ ts_phrase('rts') OR message @@ ts_phrase('panic'))
AND ts BETWEEN … ORDER BY score DESC

→ R27-M0-NA-C:J11-U11 · rts panic! · score 8.382
```

Answers label each claim as **observed**, **hypothesis** or **next step**.

TRANSPARENCY PAID OFF

1. On 4.7M lines, search reported its own fallback reason
2. Diagnostics ruled out the data and the index
3. Root cause: driver parameter binding; one-line fix, BM25 restored

Reproducible from a clean checkout in about 3 min of loading.

LIMITS

What the data does and does not establish

Establishes

- Measured line counts over the whole corpus
- Spread across nodes and days
- Owning team for each component
- What happened around an event

Does not establish

- Confirmed root causes
- Incidents: matching lines are not deduplicated
- Downtime, cost or customer impact
- Labelled anomalies for OpenStack (keyword rate)

Question → evidence → recommendation, per role. Every number checkable.